

Ground tennis-ball return: 75 mm wide-gripper candidate

Date: 2026-09-13. Latest complete metrics matrix: wide-repair-v9-screen, 11/30 complete returns under the existing evaluator tolerance and 30/30 ground lifts. The task matrix still fails. Intermediate v7 reached 9/30; historical v6 reached 4/30. The independent wide-repair-v9 re-run now has 32/32 declared videos fully decoded and verified; its positive-task labels reproduce 11/30, not 32 task successes. This is not strict nominal joint-limit qualification, PPO training success, real-time deployment validation, or hardware release.

v9 repair, verification, and remaining failures

Check	Complete metrics matrix
Positive tasks	11/30
Nominal / small / large	5/10 / 5/10 / 1/10
Real ground lifts	30/30
Open-jaw / hold negative controls	0 false successes
Fully decoded videos	32/32; 11 positive-task successes
Related code regressions	372 passed, 10 warnings
New PPO updates	0; reward/loss are not applicable, not fabricated
Hardware or manufacturing qualification	None

Successful seeds: nominal **0, 2, 3, 5, 6**; small **0, 2, 3, 6, 8**; large **8**. The other 19 episodes contain **10 robot/bin collisions, 4 timeouts, 4 premature grasp losses, and 1 non-pad robot/ball collision**. Successes increased relative to v6, but bin collisions increased from 3 to 10. **Higher completion rate is not an across-the-board safety improvement**. This remains an unreleased candidate.

Implemented changes

1. Preserve the original leg policy until a movement command produces less than 5 mm of measured progress for 2.5 s. Recovery adds a bounded 0.06 rad, 1 Hz hip roll motion. The existing 6 rad/s leg command slew limit remains unchanged.
2. Before lowering, forecast 25, 30, 35, and 40 mm crouch candidates on independent simulator copies. Authorize a docking posture only after a copy completes the entire placement, release, retreat, and stable-hold acceptance sequence.
3. Compare legacy release with a held-arm, active-leg, 0.025 rad/s slow-opening alternative. If neither predicts success, do not authorize jaw opening; never label a least-bad failing candidate as successful.
4. Slow opening screens both measured arm pose and the effective held command pose, plus jaw sweep, bottom support, whole-ball containment and prohibited contacts. Forecast results cannot overwrite the live episode's success flag.

This is a privileged simulator-truth clone teacher, not newly learned PPO or a validated real-time MPC deployment. Each docking candidate forecasts at most 2000 control steps, and each release candidate at most 800. Offline computation time is not evidence that the robot can run this planner at 50 Hz. Metadata states forecast_source=simulator_truth_clone_teacher, ppo=false, and hardware_eligible=false. Forecasts omit the renderer and independently copy model, dynamics state, and policy history. They never move live free roots or inject external forces.

The coordinated controller still commands **15 actuators: 10 legs, 4 arm-pose joints, and 1 gripper**, with no waist actuator. This is teacher/control verification, not completed whole-body PPO training.

All three MJCF files are **byte-identical to v6**, and frozen v1/v1-C sources are unchanged. Ball, bin, contacts, mass, actuator force limits and task gates were not relaxed. The 80 s horizon does not satisfy the old 15 s target. The inherited joint gate still permits 0.08 rad; the matrix measured a maximum excursion of approximately **0.012744 rad** beyond nominal ranges. Strict nominal limits, electrical/thermal/inertial measurements, the 58 g payload class, and real-time deployment remain unqualified.

v9 evidence locations

- Complete metrics: artifacts/ microduck- arm- v1c/ tennis- return/ wide- repair- v9- screen/ evaluation.json
- Per-episode result.json, telemetry.jsonl, and source-snapshot/ share that root.
- Regression log: artifacts/microduck-arm-v1c/tennis-return/repair-development/regression-final-v9.log
- Full-video rerun: artifacts/microduck-arm-v1c/tennis-return/wide-repair-v9/
- Video verification manifest: artifacts/ microduck- arm- v1c/ tennis- return/ wide- repair- v9/ video- verification/manifest.json
- Video contact sheet: artifacts/microduck-arm-v1c/tennis-return/wide-repair-v9/video-verification/contact-sheet.jpg
- Terminal-frame review sheet: artifacts/ microduck- arm- v1c/ tennis- return/ wide- repair- v9/ video- verification/final-frames-contact-sheet.jpg
- Video status: **32/32 declared videos passed hash, terminal-telemetry, frame-count, and full-decode checks; 11/30 positive tasks are successes.**

The final-frame visual review agrees with the recorded labels: all 11 pass-labeled positive episodes show the ball resting in the bin, while the remaining positives and both controls end in visibly non-pass states. This qualitative review does not relabel any episode or convert video integrity into task acceptance.

Release-profile boundary after the v9 freeze

The verified v9 bundles remain historical evidence for the **strict bottom-supported release** contract documented below. A separately identified half_height release profile is now under development: it may release from approximately half the bin height and allow a natural drop, while still requiring final whole-ball containment, settling on the bin bottom, and robot retreat. **No half_height result or video has been produced or verified yet**, and it is not included in the v9 11/30 result. Because the active specification is changing, no working-tree specification hash is used as a comparison claim against frozen v9; frozen source snapshots remain the historical comparison boundary.

An empty --video-seeds deliberately produces a metrics_only bundle, which the video verifier rejects. Explicit seeds produce metrics_and_video evidence; the verifier requires every declared seed for all three variants plus both controls, including hashes, terminal telemetry agreement, and full video decoding. Historical reports without video_seeds retain the five-video seed-0 contract.

Foot-clearance projection, velocity-based braking, always-on hip sway, and short horizon navigation prediction have separate diagnostic records. Their individual successful trials are not promoted into this matrix or a general repair claim.

Design decision

The old jaws are not permanently fixed. A separate, replaceable wide candidate now has a 75 mm open inner-pad gap, while the original 19 mm jaws and all frozen v1-C source, models and policy artifacts remain unchanged.

The experiment uses a 67 mm, 58 g nominal ball and two Type 1/2 boundary pairs: 65.4 mm / 56 g and 68.6 mm / 59.4 g. These pairs are not a full Cartesian tolerance matrix and exclude larger Type 3 balls. The experiment specification records the ITF Appendix I source. Actual felt friction, compliance and inertia are unmeasured.

Parameter	Original	Wide candidate
Open inner-pad gap	19 mm	75 mm
Gear/finger shaft spacing	24 mm	24 mm, unchanged
Outward pad offset per side	0	28 mm
Pad-center local x	23 mm	48 mm
TCP local x	30 mm	55 mm
Contact-pad size	24 × 5 × 8 mm	unchanged
Arm actuators	5	5

Parameter	Original	Wide candidate
Whole robot actuators	15	15
Link-plus-tool geometric length	135 mm	160 mm

There are ten leg actuators, four arm-pose actuators and one gripper actuator; there is no waist actuator. The original mirrored jaw coupling and motor envelopes remain intact. The width comes from offset fingers, not an impossible change in gear centers with unchanged gears.

Widening alone would let a ball centered at the old 30 mm TCP intersect the palm or motor. Moving the TCP forward to 55 mm avoids that. The supporting stems were also moved outward to ensure the pads contact the ball before the rigid stems.

Added structural blank volume corresponds to approximately 4.20 g at an assumed PA12 density of 1100 kg/m³. This excludes unresolved horn/shaft attachment, retention and fasteners. The OpenSCAD and SVG/PDF are review-only geometry, not manufacturing drawings with validated holes, fits, tolerances or strength.

Closing-angle interpretation

For simplified symmetric straight fingers with 40 mm effective length:

$$\theta = \arcsin \frac{75 - D}{2 \times 40}$$

A 66 mm ball gives approximately 6.46° per finger, or 12.92° relative closure. This approximation must not be used directly as the offset-finger servo command. For the actual candidate, a planar face-distance approximation in metres is:

$$0.0255 + 0.012 \cos \theta - 0.055 \sin \theta = D/2$$

With the ball centered at the TCP and no deformation, a 66 mm ball gives about 4.65° per finger. A 0.001 rad MuJoCo kinematic sweep measured first bilateral pad intersections at 4.98° / 4.18° / 3.32° for 65.4 / 67.0 / 68.6 mm balls. All three open snapshots are clear of ball/robot intersections; only the pads touch at first bilateral closure. **This is not dynamic grasp or force validation.** The proposed ±33 mm contact locations describe an equatorial grasp on a 66 mm ball, not the shaft spacing.

Task and acceptance

The ball starts on the floor. Both ball and robot retain free joints; no weld, suction, externally injected force or raised ball is used. The classroom bin is anchored, with 180 × 180 mm inner dimensions, 100 mm walls and 4 mm bottom/wall thickness. Its center is roughly 0.4 m from the robot's origin. This is not a validated full-height household bin.

Sequence: approach → bilateral grasp → lift → carry → lower → bottom-supported release → withdraw → remain stable.

- The initial ball must be on the floor outside the bin; spawning in the bin cannot count as a return.
- Lift the ball bottom at least 20 mm, maintaining bilateral grasp for 0.5 s.
- Carry at least 120 mm horizontally.
- The entire ball must fit inside the bin, not just its center; rim support does not count.
- Establish bottom support before releasing, with ball speed at most 0.03 m/s. Throwing and premature release fail.
- Withdraw beyond the ball radius plus 20 mm, with no robot/ball contact; retain the ball stably on the bottom for 2 s.
- Reject falls, self-collision, robot/bin collisions, joint violations and non-finite telemetry.

The ball uses analytical thin-shell inertia and uncalibrated rigid contact. Its actual mass is modeled in this separate experiment; the old small-object scenario’s 50 g construction limit was not changed.

Frozen v6 results and limitations

The evaluation is complete. These numbers describe only the frozen wide-repair-v6 bundle. Predictive-navigation / MPC and SOUTH-route experiments are not integrated into this bundle and have not established a fully qualified improvement. Source/controller review continues; this is a frozen historical measurement, not a working-tree performance guarantee or release approval.

Check	Frozen v6 result
Wide-gripper geometry at three ball sizes	3/3
Complete returns under the existing tolerance gate	4/30
Ground lifts	30/30
Nominal / small / large complete returns	4/10 / 0/10 / 0/10
Open-jaw and hold negative controls	0 false successes
Fresh verified videos	5; only nominal-0 is a task success
PPO updates for this repair	0 steps
Hardware qualification	None

The successful records are nominal seeds **0, 2, 3 and 5**, completing at 52.846, 51.650, 54.154 and 53.988 s respectively. Only seed 0 has a video; the other three successes are result/telemetry records, not additional videos. The evaluation horizon is 4000 control steps, or 80 s at 50 Hz; v2 used 750 steps, or 15 s. These completions **do not meet the old 15 s deadline**. The physical acceptance rules above are unchanged: ground start, bilateral grasp and lift, 120 mm carry, whole-ball containment, bottom-supported slow release, withdrawal, two-second stability and all safety rejections still apply. The bundle remains all_tasks_passed: false.

The 26 failures are 18 timeout, 3 robot_bin_collision, 3 premature_release_or_throw, 1 non_pad_ball_collision, and 1 unsupported_or_fast_release. Both negative controls time out without success.

Joint-limit review caveat: the evaluator already allowed a **0.08 rad** joint-limit tolerance; neither v5 nor v6 introduced it. Review found that successful nominal episodes exceed the nominal 1.8 rad elbow limit by approximately 0.009 - 0.011 rad under MuJoCo soft limits. Thus 4/30 passes the existing legacy tolerance gate, **not strict nominal joint-limit qualification**. Do not infer that every physical constraint is perfectly satisfied or that hardware is safe. V6 now records maximum_nominal_joint_excursion_rad on each telemetry frame, alongside evaluator_joint_tolerance_rad, making nominal-limit overshoot explicit without tightening the legacy acceptance gate.

The robot has 15 actuators: 10 legs, 4 arm-pose joints and 1 gripper, with 0 waist actuators. tennis_controller.py adds a coordinated whole-body kinematic teacher around the legacy alpha_walking.onnx leg policy. It uses actual-state feedback for crouching, downward ground IK, a folded loaded lift, loaded navigation and collision/torque-screened placement. This is not a newly trained end-to-end policy and not new PPO.

V6 retains the v5 placement crouch of **25 mm** and caches the last safe release IK target if the new target is unreachable or its shoulder pitch is **≥ 0.62 rad**. In the blocked-plan fallback, the gripper opens after at least **1 s**, after ball speed has remained **< 0.001 m/s for 0.3 s**, and only after the additional cached-pose release checks pass.

V6 formalizes the blocked-plan stage as release_cached_pose. Before opening from the cached pose, it requires actual bottom support, whole-ball containment inside the bin and no forbidden contacts, plus a **nine-position jaw-opening collision sweep on scratch data**, not the live robot. The sweep allows intended pad/ball contacts and rejects other penetrating arm contacts. This discrete kinematic screen is not a continuous-motion or hardware safety proof. release_plan_blocked and cached_release_screen_passed are now explicit telemetry fields; the unrecorded fallback limitation belongs to historical v5.

The physical monitor still checks bottom support, release speed, containment, withdrawal and stability. Its acceptance thresholds and the motor limits are unchanged. The nominal, small and large scene XML files have identical SHA-256 hashes across v2, v5 and v6; this verifies unchanged scene files, not hardware qualification or every source dependency. This frozen run is not a release or a claim that all constraints are perfect.

The frozen regression log reports **347 passed, 10 warnings**. It covers the repair while retaining the acceptance defenses, including rejection of false initial-bin success and unsafe release behavior. Passing tests does not convert the 4/30 matrix into task acceptance or strict joint-limit qualification.

The ball alone produces approximately 0.091 N · m at a 160 mm horizontal lever arm, before arm, fingers, acceleration and safety margin. The old 0.1 N · m simulation screening limit is not a qualified continuous actuator rating.

Next engineering gates remain loaded wide-jaw bench testing, collision-free wrist orientation and whole-body crouch/reach, reliable loaded navigation and placement, then demonstrations and whole-body learning. This experiment has **zero new PPO training steps** and no task-qualified tennis-return ONNX export. Old 10 g task exports do not prove a 58 g tennis-ball capability. Nothing in v6 was run on hardware.

Historical v5, v4 and v2 results

Keep earlier evidence for comparison, not as a description of v6:

- Frozen wide-repair-v5: 4/30 complete returns under the same legacy joint-limit tolerance, 30/30 ground lifts, 4000 steps / 80 s. Successes are nominal seeds 0, 2, 3 and 5. Its five verified videos contain only one success (nominal-0); its regression log records 302 passed. The blocked-plan cached release was not explicitly recorded in telemetry and lacked v6' s formal release screen.
- Frozen wide-repair-v4: 1/30 complete returns, 30/30 ground lifts, 4000 steps / 80 s; nominal seed 0 completes at 56.228 s. Its five verified videos contain one success and four failures. Its regression log records 297 passed.
- wide-screen-v2 completed 0/30 tasks.
- Its five videos (nominal-0, small-0, large-0, open_jaw, hold) are all failure videos.
- That run used the earlier 750-step / 15 s horizon and predates the coordinated v4 crouch, folded-lift, loaded-navigation and placement controller.

Keep the old results separate

The original teacher matrix now passes 120/120 free-base and 80/80 fixture episodes. However, the actual teacher-plus-whole-body-residual ONNX matrix passed **100/120**, not 120/120. Remaining failures are primarily stowed walking and walking carry. Those models use 69 observations and 15 residual outputs and retain their teacher. This candidate does not modify those frozen artifacts or claim independent learned competence.

Artifacts and reproduction

- Sources: `microduck_arm_experiments/ tennis_return.py` and `microduck_arm_experiments/ tennis_controller.py`
- Specification: `hardware/microduck-arm-v1c/experiments/tennis-return.json`
- CAD/dimensions: `hardware/microduck-arm-v1c/experiments/tennis-wide-gripper/`
- Stock results: `artifacts/microduck-arm-v1c/tennis-return/stock-screen-v2/evaluation.json`
- Historical wide result: `artifacts/microduck-arm-v1c/tennis-return/wide-screen-v2/evaluation.json`
- Historical v4 result: `artifacts/microduck-arm-v1c/tennis-return/wide-repair-v4/evaluation.json`
- Historical v5 result: `artifacts/microduck-arm-v1c/tennis-return/wide-repair-v5/evaluation.json`
- Frozen v6 result: `artifacts/microduck-arm-v1c/tennis-return/wide-repair-v6/evaluation.json`
- Frozen v6 regression: `artifacts/microduck-arm-v1c/tennis-return/repair-development/regression-final-v6.log`

- Frozen v6 media verification and provenance hashes: artifacts/microduck-arm-v1c/tennis-return/wide-repair-v6/video-verification/manifest.json

The five exact frozen v6 video paths are:

- artifacts/microduck-arm-v1c/tennis-return/wide-repair-v6/nominal-0/rollout.mp4
- artifacts/microduck-arm-v1c/tennis-return/wide-repair-v6/small-0/rollout.mp4
- artifacts/microduck-arm-v1c/tennis-return/wide-repair-v6/large-0/rollout.mp4
- artifacts/microduck-arm-v1c/tennis-return/wide-repair-v6/open_jaw/rollout.mp4
- artifacts/microduck-arm-v1c/tennis-return/wide-repair-v6/hold/rollout.mp4

The verifier fully decodes exactly those five videos, checks frame counts against telemetry, requires each result.json to equal its evaluation record, and requires the terminal telemetry outcome to equal the recorded success and failure fields. It does not inspect a failed video and relabel it as successful. The manifest records one verified success (nominal-0) and four verified failures.

The manifest retains full SHA-256 hashes for the evaluation, episode results, telemetry, videos, source snapshots, verifier and contact sheet. It establishes internal hash consistency, not cryptographic attestation, independent physics re-evaluation or strict joint-limit qualification.

Exact v6 SHA-256 values from that manifest:

- Evaluation: 91a6da825f7d73ec21b0daf1cb01d7659c16cf6bda3b4474cbe59e8031c0597c
- Controller snapshot: fa8cad9e55ac71fc8b99d8e179c62a08c266498b56f1d1ec7445f970f65e33a3
- Recorded verifier: 15a2bb0c32e1b4fc64e5d84b4f91ad49fba376ff789b99301380ec01fc2e79c8

From the repository root, use a fresh output path. These commands run the working-tree controller; they do not promise to reproduce frozen v6 after later source changes. Compare the new source hashes with v6 before attributing results.

```
PYTHONPATH=.:microduck_local/src OMP_NUM_THREADS=1 \  
rlx/.venv-microduck/bin/python -m microduck_arm_experiments.tennis_return \  
--gripper wide_candidate --out artifacts/tennis-new-run \  
--max-steps 4000 --workers 4 --video-seeds 0
```

```
PYTHONPATH=.:microduck_local/src OMP_NUM_THREADS=1 \  
rlx/.venv-microduck/bin/python -m pytest \  
rlx/tests/test_microduck_arm_tennis_return.py -q
```

```
PYTHONPATH=.:microduck_local/src \  
rlx/.venv-microduck/bin/python scripts/verify_tennis_videos.py \  
--root artifacts/tennis-new-run
```

The evaluation CLI now exits 1 when any positive task fails or a negative control succeeds; the frozen v6 matrix therefore has a nonzero task status even though it wrote complete evidence. Run the verifier separately after the evaluation command; shell automation must account for the expected nonzero exit when collecting a partially successful diagnostic matrix. Passing unit tests or video-integrity checks is not task acceptance. On macOS, offscreen rendering needs host graphics access; omitting `--video-seeds` runs without videos.